Enterprise Knowledge Workspace

Product Design Brief: AI Workflow for Knowledge Reuse and Decision Preparation

> Version: Public English version v0.3

> Purpose: A concise product case study for the English portfolio page.

> Status: Product-grade prototype / demo-able MVP.

> Boundary: This document does not claim production deployment. It contains no private files, internal company documents, or sensitive data paths.

This case study explores how scattered organizational knowledge can become a source-grounded, reviewable, and maintainable AI workflow.

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0. How to Read This Document

This is a **Product Design Brief**, not a full PRD or engineering specification.

It is intentionally shorter than the Chinese PRD-style document. For an international product audience, the focus is on:

- problem framing;
- user jobs;
- scope and trade-offs;
- product decisions;
- workflow design;
- validation and failure cases;
- governance and scaling boundaries.

The core question is:

> How can scattered organizational knowledge be turned into a traceable, reviewable, and continuously improving AI workflow, instead of remaining a one-off document search or chatbot experience?

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1. Product Overview

1.1 Background

In many organizations, valuable knowledge does not live in one clean knowledge base. It is distributed across project documents, meeting notes, decision records, review materials, operating rules, retrospectives, customer feedback, and cross-functional communication.

As information accumulates, traditional folders, keyword search, and personal memory become increasingly insufficient:

- People remember that a similar issue was discussed before, but not where the source material lives.
- New team members spend significant time rebuilding context.
- Project owners need evidence for decisions, but the supporting materials are fragmented.
- Retrospectives are written once, then rarely become reusable organizational memory.

The product is therefore not designed as a generic Q&A bot. It is designed as a knowledge workflow:

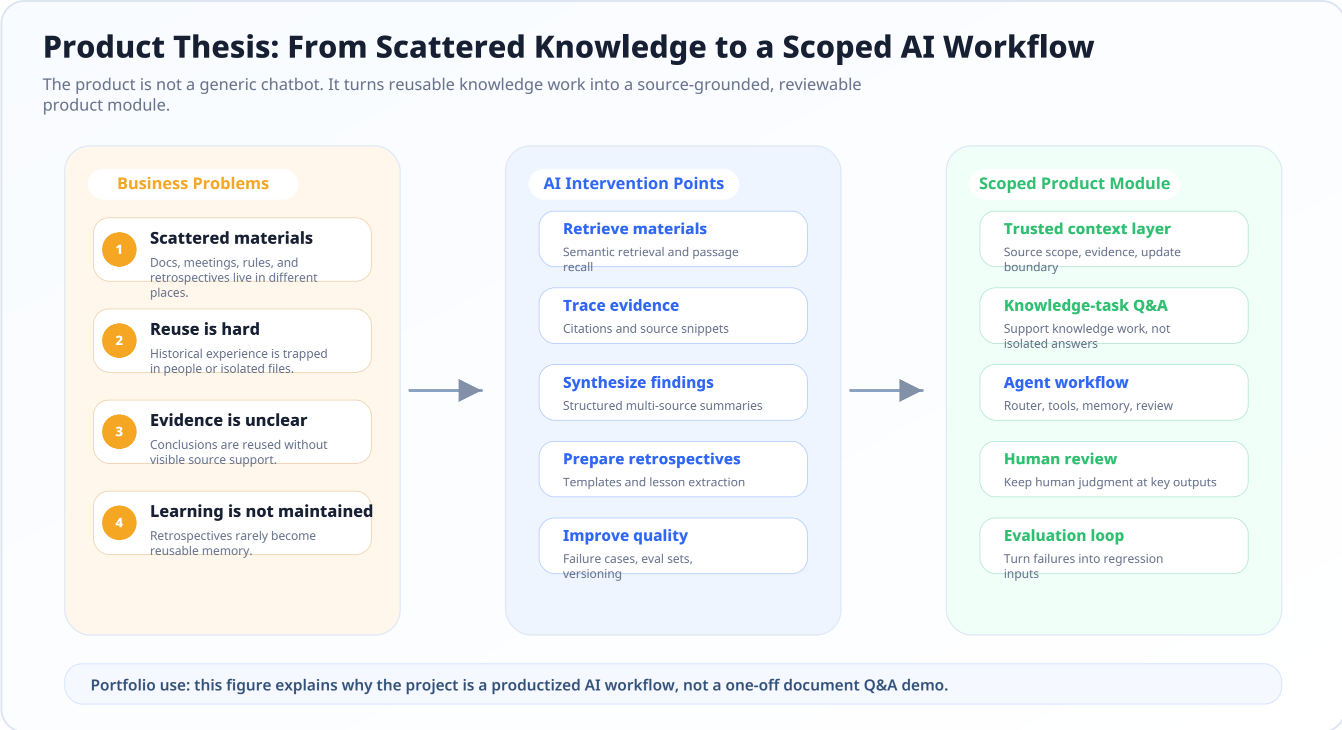
> Retrieve relevant materials -> trace evidence -> synthesize conclusions -> support project retrospectives -> extract reusable learning -> improve through failure cases and evaluation.

1.2 Product Definition

The **Enterprise Knowledge Workspace** is a scoped AI product module for knowledge reuse and decision preparation. It turns scattered documents and historical materials into a workflow that is:

- searchable;
- source-traceable;
- reviewable;
- retrospective-friendly;
- maintainable;
- and iteratively improvable.

Its value is not to let AI generate answers freely, but to help users complete knowledge work based on controlled sources and human review.



Product thesis map

2. Business Problem and User Jobs

2.1 Typical Knowledge Sources

Organizational knowledge may come from:

- project documents and phase deliverables;
- meeting notes and decision records;
- retrospective reports and lessons learned;
- business rules, process documents, and operating manuals;
- customer feedback, after-sales records, and quality issue records;
- historical cases, competitor analysis, and management reports.

The challenge is not a lack of knowledge. The challenge is the lack of a sustainable product mechanism for reusing it.

2.2 Core Business Problems

Problem	What Happens	Impact
Scattered materials	Knowledge is spread across documents, notes, records, and discussions.	Users rely on memory, keywords, and manual follow-ups.
Reuse is difficult	Historical experience remains in individuals or isolated documents.	Teams repeat research and repeat mistakes.
Evidence is hard to trace	Conclusions are reused without clear source evidence.	Decision confidence and accountability are weakened.
Retrospectives do not become assets	Lessons learned are not indexed or continuously improved.	Organizational memory remains weak.

2.3 User Roles and Jobs

User Role	Typical Need
Project owner / Product manager	Find historical decisions, project references, and supporting evidence.
New team member	Understand project context, key decisions, and relevant materials quickly.
Manager / Decision participant	Review the basis of conclusions, risks, and historical cases.
PMO / Knowledge management role	Maintain reusable knowledge and reduce repeated communication.

Users are not only searching for files. They are trying to complete knowledge tasks: find relevant materials, trace supporting evidence, synthesize conclusions, prepare retrospectives, and preserve reusable learning.

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3. AI Intervention Points

AI is not introduced only at the document Q&A layer. It is inserted into multiple points in the knowledge workflow.

Workflow Step	Traditional Pain Point	AI Intervention	Product Value
Find materials	Users depend on keywords and memory.	Semantic retrieval and related passage recall.	Lower search cost.
Trace evidence	Users still need to read many documents manually.	Source citation and evidence extraction.	Make conclusions verifiable.
Synthesize conclusions	Cross-document synthesis is time-consuming.	Multi-document summarization and structured synthesis.	Improve preparation efficiency.
Prepare retrospectives	Retrospectives rely heavily on personal memory.	Retrospective templates and lesson extraction.	Turn experience into reusable knowledge.
Improve quality	Errors are hard to track and learn from.	Failure case logs, eval sets, and version records.	Support continuous improvement.

The product interface is designed around these jobs rather than around a single chat input.

Workspace home

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4. Key Product Decision: From RAG to a Trusted Context Layer

4.1 The Model Is Stronger, But the Business Problem Is Different

As foundation models become stronger, many low-risk, general-purpose, one-off text tasks can be completed directly by the model. A complex intermediate layer is not always necessary.

Knowledge reuse is different. The key question is not whether the model can produce a fluent answer. The key questions are:

- Does the answer come from the specified source scope?
- Can the user trace it back to original evidence?
- Can the knowledge be updated as internal materials change?
- Can access and usage boundaries be controlled?
- Can human users review and approve important outputs?
- Can errors be logged, evaluated, and improved over time?

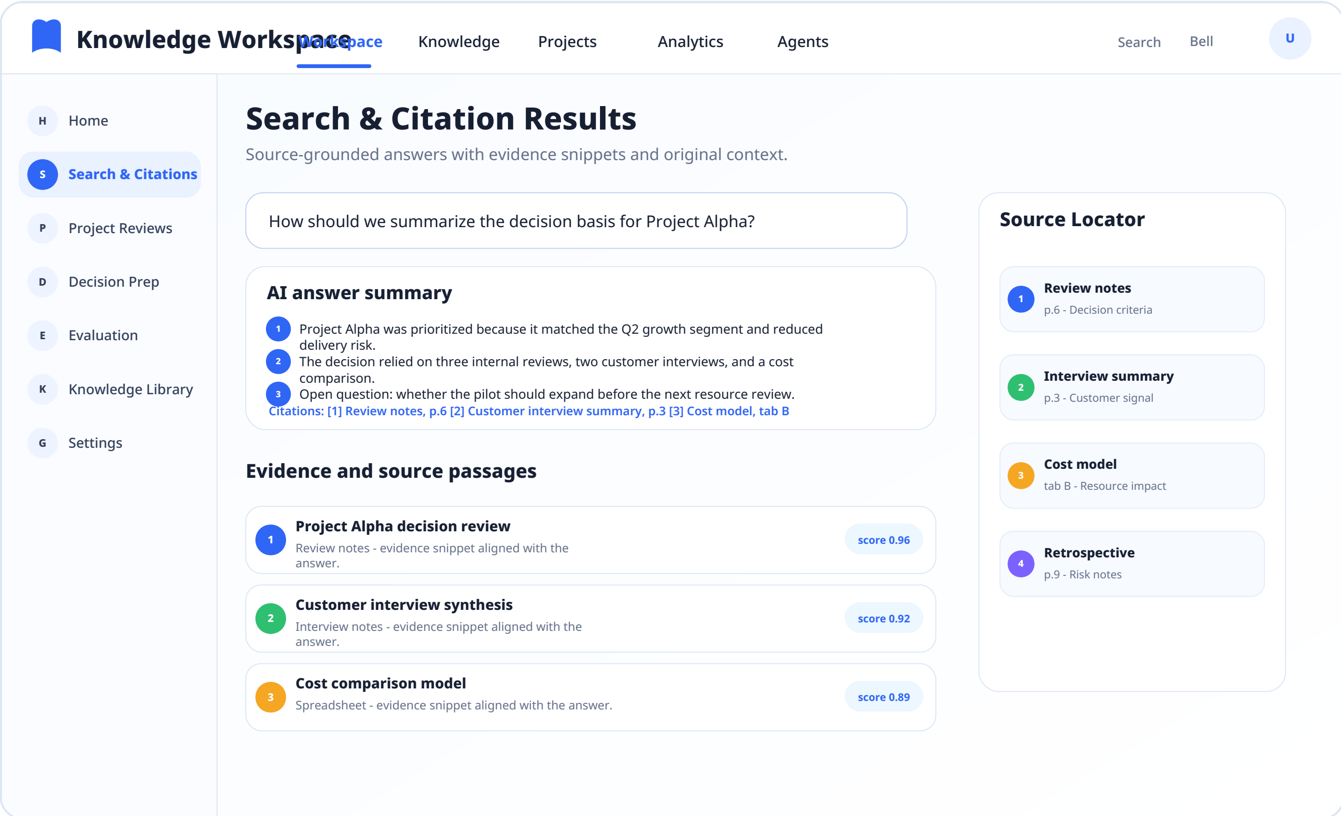
Therefore, the system needs a **trusted context layer** between the foundation model and internal knowledge assets.

4.2 RAG as Context Infrastructure

In this project, RAG is not positioned as the final product value or a technical selling point. It is treated as one implementation of a trusted context layer.

Its role is to provide controlled source scope, source traceability, knowledge update mechanisms, reviewable evidence, organizational memory, and quality improvement through failure cases and evaluation.

> The value of the context layer is not to make the model smarter. It is to make model outputs usable inside organizational knowledge workflows, where source, boundary, evidence, reviewability, evaluation, and rollback all matter.



Search and citation results

4.3 Direct Model Call vs. Trusted Context Layer

Question	Direct Foundation Model Call	Trusted Context Layer
Internal knowledge	Cannot guarantee coverage of private or updated materials.	Uses specified internal materials.
Evidence	Source evidence is difficult to guarantee.	Provides traceable references.
Knowledge updates	Depends on model updates or manually supplied context.	Can update with the knowledge source.
Access boundaries	Hard to manage at document or domain level.	Can be designed around source scope and permissions.
Verification	Output may be plausible but hard to verify.	Users can return to original evidence.

5. Product Route and Trade-offs

5.1 Why Not Just Use a Foundation Model?

A foundation model is useful for general explanation, writing, and synthesis. But it does not naturally know the organization's latest private materials, and it cannot consistently provide source-grounded evidence unless a source mechanism is designed.

Product decision:

> The foundation model can serve as the reasoning and language layer, but not as the source-of-truth layer for organizational knowledge.

5.2 Why Not Prioritize Fine-Tuning?

Fine-tuning may help with stable classification, tone, or output style. But it does not solve the core product problem here: source freshness, evidence traceability, and knowledge update.

Product decision:

> Fine-tuning is not the first priority for this product stage. Source-grounded context and workflow control matter more.

5.3 Product Evolution

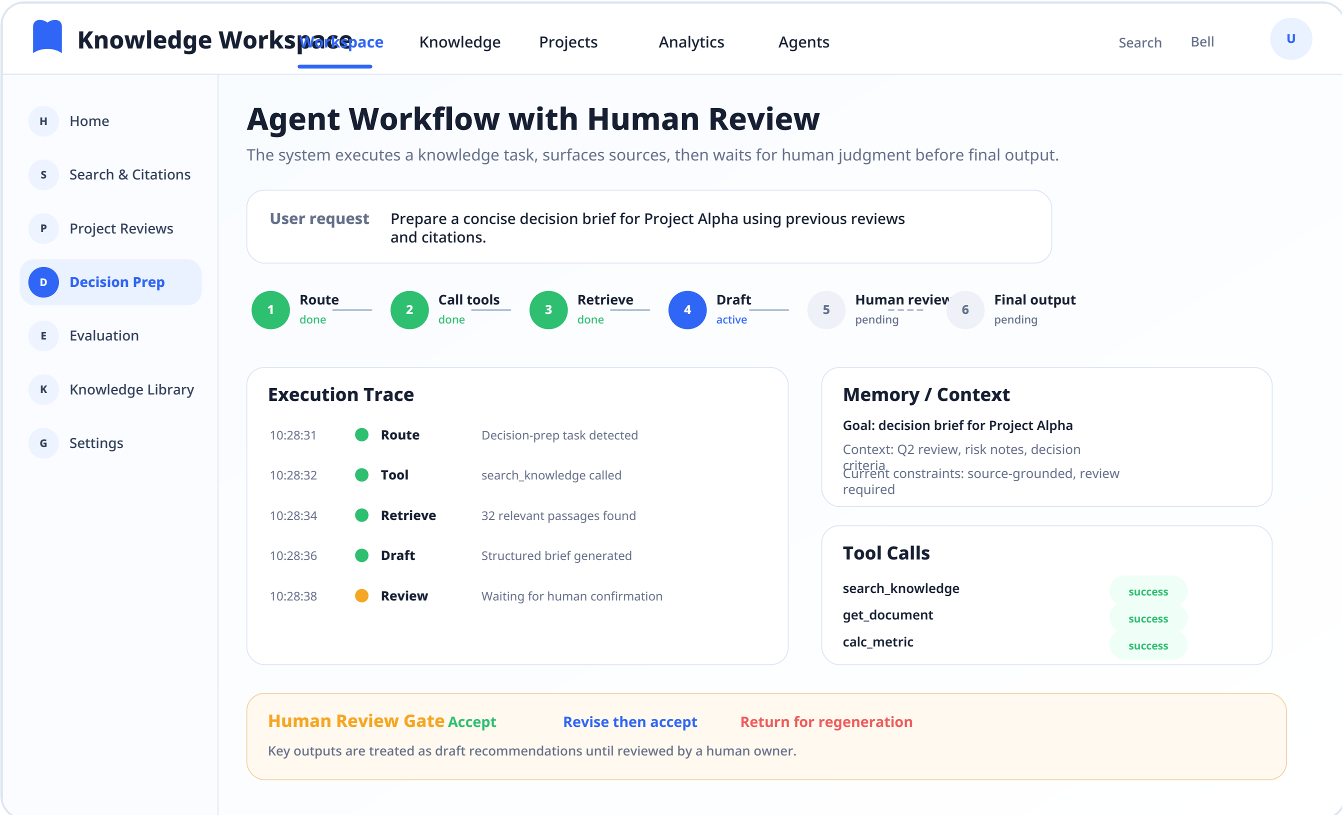
Product evolution

Stage	Core Question	Product Decision
V1: Trusted Retrieval	Can users find source-grounded answers?	First solve retrieval, citation, and verification.
V2: Agent Workflow	Can users complete knowledge tasks?	Add routing, tools, memory, and human review.
V3: Maintainable MVP	Can the workflow improve over time?	Add eval sets, failure case logs, versioning, and rollback.

6. Agent Workflow Design

The Agent layer is introduced only after the trusted context layer is established. The goal is not to make the system act autonomously. The goal is to help users complete structured knowledge work with visibility and control.

Component	Product Role
Router	Decide whether the user is searching, preparing a retrospective, drafting a summary, or needs clarification.
Tool calls	Convert retrieval, summarization, retrospective support, and metric lookup into callable capabilities.
Trusted context layer	Provide source-grounded evidence for the workflow.
Memory	Keep short-term task context and current goals.
Execution trace	Show why the system retrieved sources, called tools, and produced the draft.
Human review	Keep human judgment before key outputs are accepted or reused.



Agent workflow with human review

7. Validation and Failure Case Loop

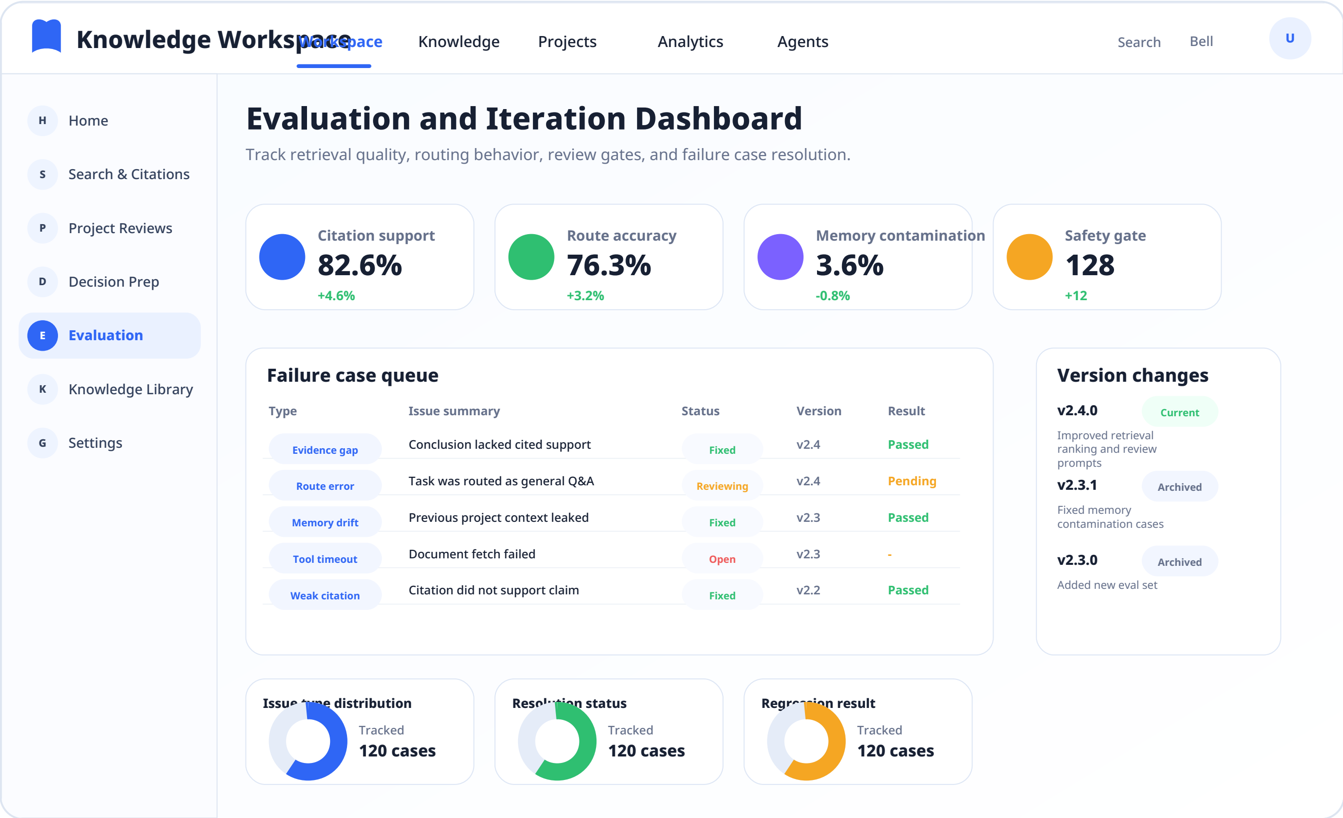
This project is evaluated not only by whether it can generate an answer. It is evaluated by whether the answer is source-grounded, reviewable, and improvable.

7.1 Evaluation Dimensions

Dimension	What to Validate
Source recall quality	Does the correct source appear in the retrieved set?
Citation support rate	Does the cited passage actually support the claim?
Route accuracy	Does the Router choose the right workflow path?
Memory contamination rate	Does prior context incorrectly affect a new topic?
Trace completeness	Can the user see route, tools, sources, and output basis?
Safety gate effectiveness	Are write, delete, or publish-like actions gated by human review?

7.2 Representative Failure Cases

Failure Case	What It Reveals	Iteration Direction
Similar but wrong source retrieved	Semantic similarity is not the same as business relevance.	Add project/domain filters and citation review.
Fluent answer without sufficient evidence	The model may over-answer when source support is weak.	Add insufficient-evidence responses and citation checks.
Router sends a knowledge task to direct Q&A	Task classification affects user trust and output quality.	Improve route rules and add regression cases.



Evaluation dashboard

8. Human Review, Governance, and Scaling Considerations

8.1 Human-AI Collaboration Principles

- AI retrieves, organizes, drafts, and highlights evidence.
- Humans confirm conclusions, make trade-offs, and approve reusable outputs.
- High-impact outputs must remain reviewable and traceable.
- The system should not automatically write back, delete, publish, or make final decisions.

8.2 Current Boundary

The current prototype does not include:

- full enterprise permission systems;
- multi-user governance workflows;
- automatic write-back to company systems;
- external publishing;
- replacement of human judgment;
- production deployment claims.

8.3 Scaling Considerations

To move toward a production-grade product, the next layer would include:

- permission design;
- audit logs;
- sensitive information handling;
- multi-user collaboration;
- system integrations;
- evaluation dashboards;
- versioning and rollback policies;
- maintenance ownership for knowledge assets.

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9. Relationship with the AI Requirement Flow Workspace

The Enterprise Knowledge Workspace focuses on **historical knowledge, evidence, retrospectives, and decision preparation**.

The AI Requirement Flow Workspace focuses on **current business signals, requests, issues, risk cards, responsibility routing, and status tracking**.

Module	Core Object	Key Question
Enterprise Knowledge Workspace	Knowledge assets, historical experience, decision evidence	How can past knowledge be retrieved, cited, reviewed, and reused?
AI Requirement Flow Workspace	Current requirements, issues, risks, ownership, and status	How can new business signals be identified, confirmed, routed, tracked, and reviewed?

This boundary keeps the knowledge workspace from becoming a task-management system, while still allowing it to generate evidence-based judgment references.

Appendix: What Is Intentionally Omitted

The public version intentionally omits:

- full implementation plans;
- detailed API schemas;
- raw test cases and full failure logs;
- private file paths or source names;
- interview scripts and internal portfolio notes.

These materials remain in the internal working version for deeper interview preparation and future project iteration. They are not necessary for a public product design brief.